

Show all steps. Simplify answers fully. Write answers using positive exponents.

1. Write 4 750 000 in scientific notation.

2. Evaluate: 3^4

3. Find 18% of 450.

4. Classify each number as natural, whole, integer, rational, or irrational.

	Natural	Whole	Integer	Rational	Irrational
7					
0					
-12					
$\frac{3}{5}$					
$\sqrt{7}$					

5. Evaluate: $8 + 3(6 - 2)$

6. Solve for x : $5x - 7 = 28$

7. Solve for b : $A = bh$

8. Simplify. x^4x^7

9. Simplify: x^0

10. Simplify and write using positive exponents: $\frac{6x^5y^{-2}}{3x^{-1}y^3}$

11. Is the number rational or irrational? $\sqrt{49}$

12. Write 0.0000832 in scientific notation.

13. Evaluate: $(-2)^5$

14. What percent of 250 is 65?

15. Evaluate: $24 \div 6 + 3(5)$

16. Solve for x : $3(x + 4) = 27$

17. Solve for t : $d = rt$

18. Is the number rational or irrational? 0.375

19. Simplify. $\frac{a^9}{a^4}$

20. Write using positive exponents: x^{-5}

21. A laptop originally costs \$1 250. Its price is reduced by 12%. What is the sale price?

22. Evaluate: $3(-2)^3 + 5(4 - 1)$

23. Write 6.14×10^5 in standard notation.

24. Evaluate: -4^2

25. 42 is 35% of what number?

26. Solve the inequality: $6 - 4x < 2x + 18$

27. Is the number rational or irrational? π

28. Evaluate: $5 + 2^3(4 - 1)$

29. Solve for x : $7x + 9 = 4x + 30$

30. Solve for w : $P = 2l + 2w$

31. Simplify: $(m^3)^4$

32. Write using positive exponents: $\frac{1}{a^{-3}}$

33. Solve: $5(2x - 3) - 4 = 3(x + 2) + 10$

34. Evaluate: $6 + 2[5^2 - 3(4 + 2)]$

35. Write 3.07×10^{-4} in standard notation.

36. Evaluate: $18 - 3(2 + 4) + 5$

37. Solve the inequality: $2(3x - 1) \leq 4x + 10$

38. Solve for x : $4(2x - 3) = 5x + 15$

39. Simplify. $(2x^3)^2$

40. Evaluate: $5[3 + (8 - 2)^2] \div 3$

41. Solve for x : $3(2x - 5) + 4 = 2(x + 7) + 7$